Test Case 1 (Normal Input)

```
========== MENU ==========
1. Employee Summary
2. Department-wise Analysis and Salary Filter
3. Show All Employees
4. Show Department-wise Bar Chart
5. Exit
==========================
Enter your choice: 1

========== EMPLOYEE SUMMARY ==========
Employees: 3
Average salary: 53333.33
Highest paid: Karan (70000)
Total payroll: 160000
======================================
```

Text Case 2 (Minimum Input)

```
========== MENU ==========
1. Employee Summary
2. Department-wise Analysis and Salary Filter
3. Show All Employees
4. Show Department-wise Bar Chart
5. Exit
==========================
Enter your choice: 1

========== EMPLOYEE SUMMARY ==========
Employees: 1
Average salary: 50000.00
Highest paid: Aman (50000)
Total payroll: 50000
======================================
```

Test Case 3 (Invalid Input)

```
----------------------------------------
       EMPLOYEE DATA ANALYZER
========================================

Original Data:
  Emp_ID   Name Department   Salary  Experience
0     E1   Aman         IT  50000.0           3
1     E1   Aman         IT  50000.0           3
2     E4   Neha         HR      NaN           4
3     E2   Riya         HR  40000.0           2
4     E3  Karan         IT  70000.0           5

Data cleaned successfully.
Duplicates removed: 1
Missing salaries filled with mean: 53333.33
Cleaned rows: 4
```

Test Case 4 (Different Valid Scenario)

```
========================================
       EMPLOYEE DATA ANALYZER
========================================

Original Data:
  Emp_ID   Name Department  Salary  Experience
0     E1   Aman         IT   50000           3
1     E2   Riya         HR   40000           2
2     E3  Karan         IT   70000           5

Data cleaned successfully.
Missing salaries filled with mean: 53333.33
Cleaned rows: 3


========== MENU ==========
1. Employee Summary
2. Department-wise Analysis and Salary Filter
3. Show All Employees
4. Show Department-wise Bar Chart
5. Exit
==========================
Enter your choice: 2

========== DEPARTMENT-WISE ANALYSIS ==========
HR: 1 employees | Avg salary: 40000.00
IT: 2 employees | Avg salary: 60000.00
==============================================

Enter salary limit: 45000

Employees with salary above 45000:
Aman - IT - 50000
Karan - IT - 70000


========== MENU ==========
1. Employee Summary
2. Department-wise Analysis and Salary Filter
3. Show All Employees
4. Show Department-wise Bar Chart
5. Exit
==========================
Enter your choice: 4
```

```
==========================
Enter your choice: 4

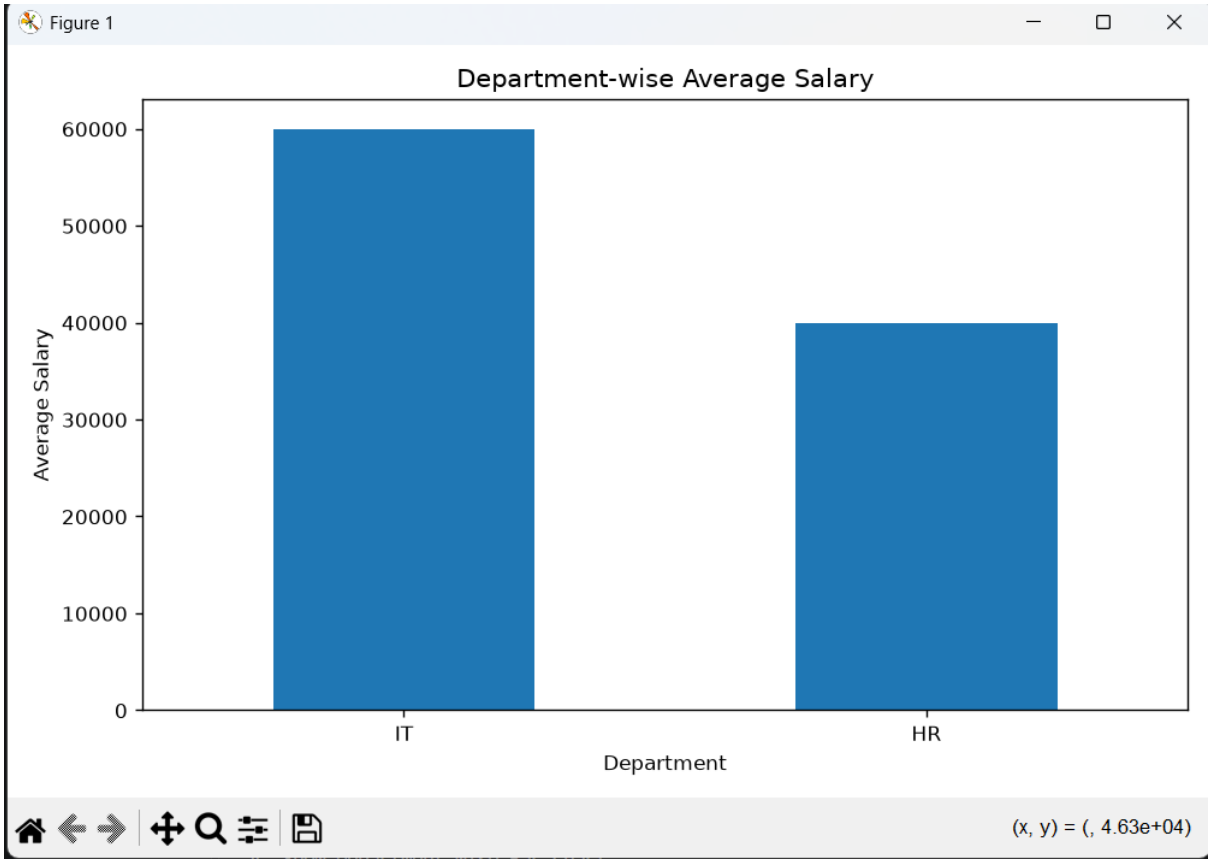

========== MENU ==========
1. Employee Summary
2. Department-wise Analysis and Salary Filter
3. Show All Employees
4. Show Department-wise Bar Chart
5. Exit
==========================
Enter your choice: 45000

Invalid choice.
Please enter a number from 1 to 5.


========== MENU ==========
1. Employee Summary
2. Department-wise Analysis and Salary Filter
3. Show All Employees
4. Show Department-wise Bar Chart
5. Exit
==========================
Enter your choice: 3

========== ALL EMPLOYEES ==========
Emp_ID  Name Department  Salary  Experience
    E1  Aman         IT   50000           3
    E2  Riya         HR   40000           2
    E3 Karan         IT   70000           5
====================================
```

bar chart


Figure 1
Department-wise Average Salary
60000
50000
40000
30000
20000
10000
0
Average Salary
IT
HR
Department
(x, y) = (, 4.63e+04)